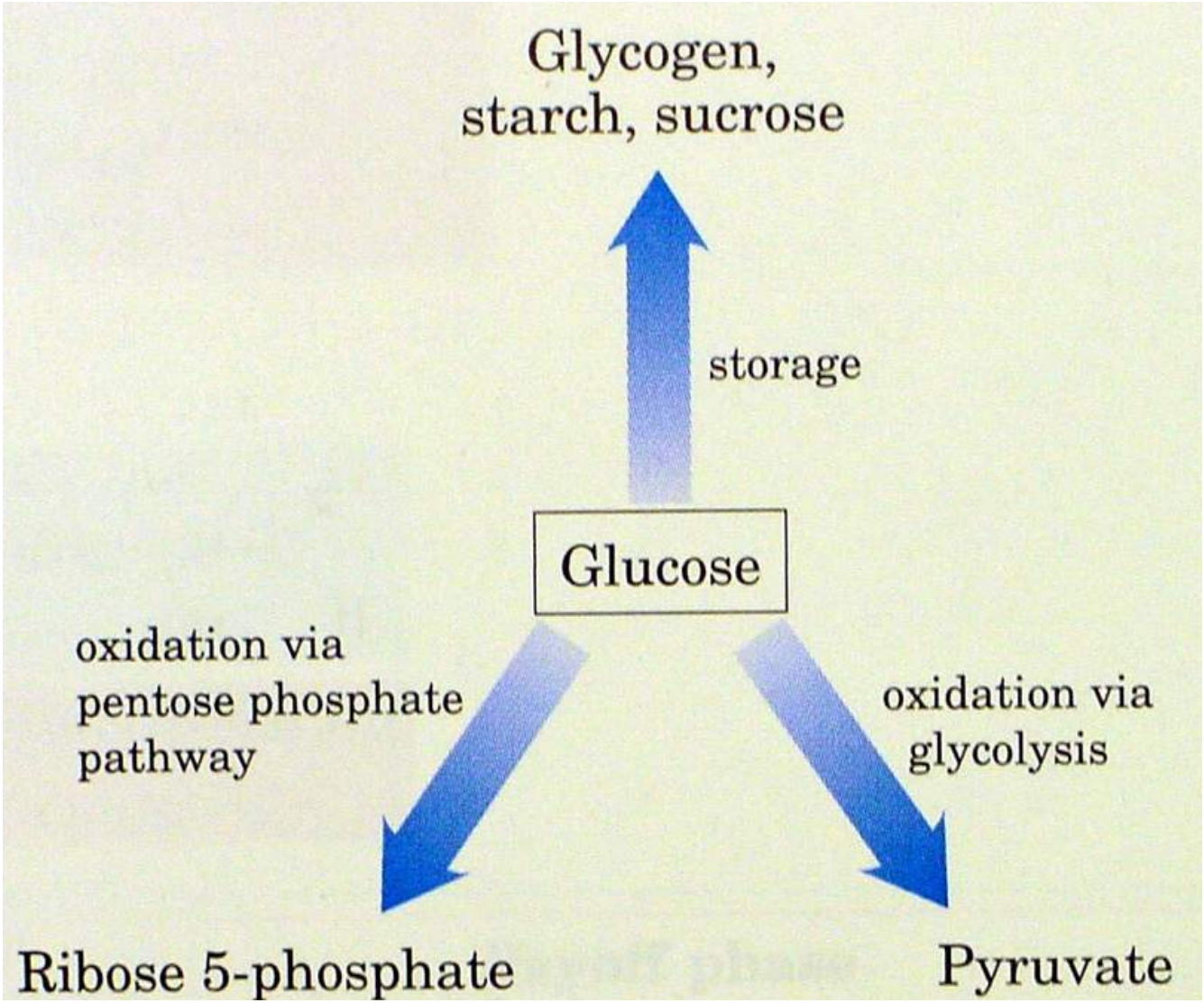
PENTOSE PHOSPHATE PATHWAY
OR
HEXOSE MONOPHOSPHATE SHUNT
OR
PHOSPHOGLUCONATE PATHWAY

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OBJECTIVES

- Overview
- Reactions
- Uses of NADPH
- To relate defects in the pentose phosphate pathway to disease conditions.

- The pentose phosphate pathway is an alternate route for the oxidation of glucose.



Introduction

- The pentose phosphate pathway occurs in the **cytosol of the cell.**
- It includes **irreversible oxidative reactions,** followed by a series of **reversible sugar–phosphate** interconversions.
- No adenosine triphosphate (ATP) is directly consumed or produced in the cycle.
- One Carbon of glucose 6-phosphate is released as CO2, and two reduced nicotinamide adenine dinucleotide phosphates (NADPHs) are produced for each glucose 6-phosphate molecule entering the oxidative part of the pathway.

cont..

- The rate and direction of the reversible reactions of the pentose phosphate pathway are determined by the supply of and demand for intermediates of the cycle.

- The pathway provides a major portion of the body's NADPH, which functions as a **biochemical reductant.**

The pentose phosphate pathway has two main functions

- **Generation of NADPH**

  - mainly used for reductive syntheses of fatty acids, steroids, amino acids; and production of reduced glutathione in erythrocytes and other cells.

  - active in liver, adipose tissue, adrenal cortex, erythrocytes, testes, and lactating mammary gland

- **Production of ribose residues for nucleotide and nucleic acid synthesis.**

1st phase

IRREVERSIBLE OXIDATIVE REACTIONS

- Glucose 6-phosphate undergoes dehydrogenation and decarboxylation to give a ribulose 5-phosphate, which is converted to its isomer, ribose 5-phosphate.

- Overall equation of 1st phase:

Glucose 6-phosphate + 2 $NADP^+$ + H_2O →

ribose 5-phosphate + CO_2 + 2 NADPH + 2 H^+

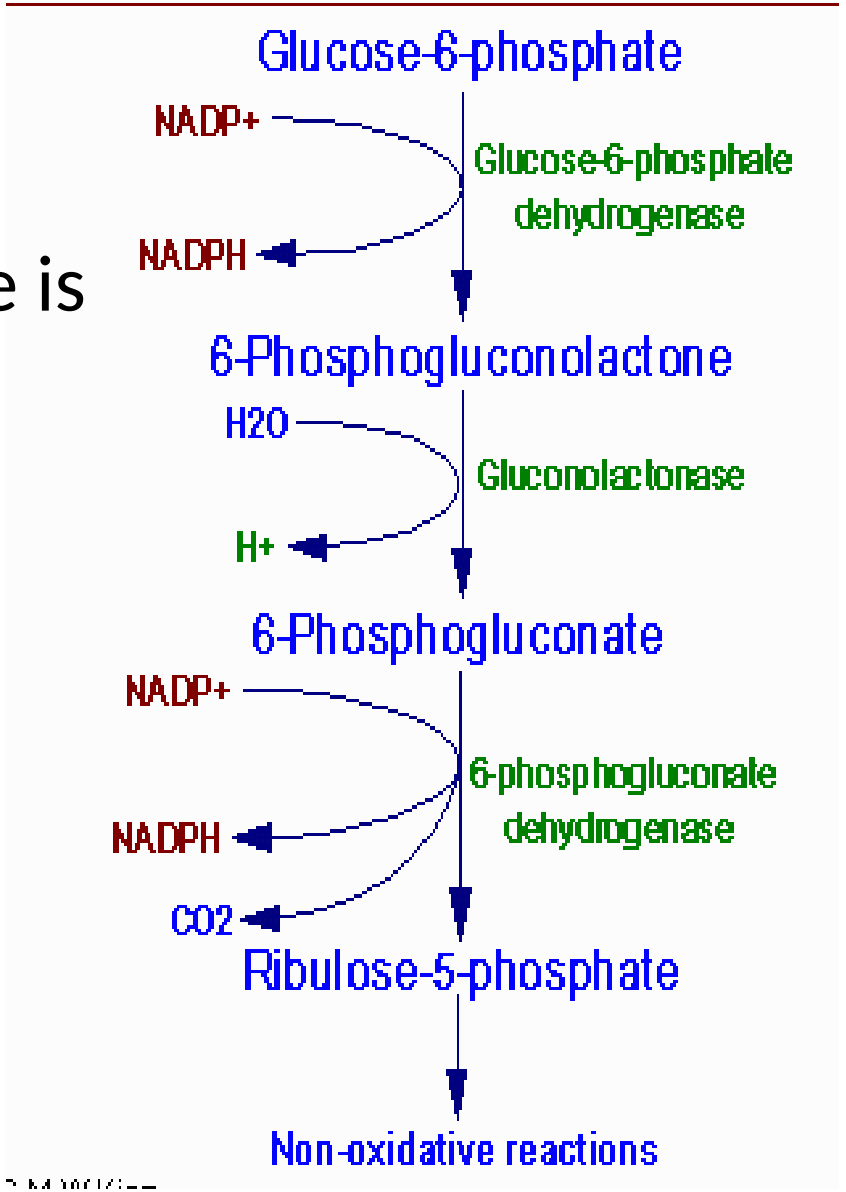
1. Dehydrogenation of glucose 6-phosphate

- Glucose 6-phosphate dehydrogenase (G6PD) catalyzes an irreversible oxidation of glucose 6-phosphate.
- Oxidized NADP (NADP+) acts as a coenzyme.
- Regulation takes place at the G6PD reaction.

2. Formation of ribulose 5-phosphate:

- 6-Phosphogluconolactone is hydrolyzed by 6 -phosphogluconolactone hydrolase.

- The oxidative decarboxylation of the product, 6-phosphogluconate, is catalyzed by 6-phosphogluconate dehydrogenase.



2nd Phase:

REVERSIBLE NONOXIDATIVE REACTIONS

- The nonoxidative reactions of the pentose phosphate pathway occur in all cell types synthesizing nucleotides and nucleic acids.

- These reactions catalyze the interconversion of sugars containing three to seven carbons.

- These reversible reactions permit ribulose 5-phosphate (produced by the oxidative portion of the pathway) to be converted either to ribose 5-phosphate (needed for nucleotide synthesis or to intermediates of glycolysis (that is, fructose 6-phosphate and glyceraldehyde 3-phosphate).

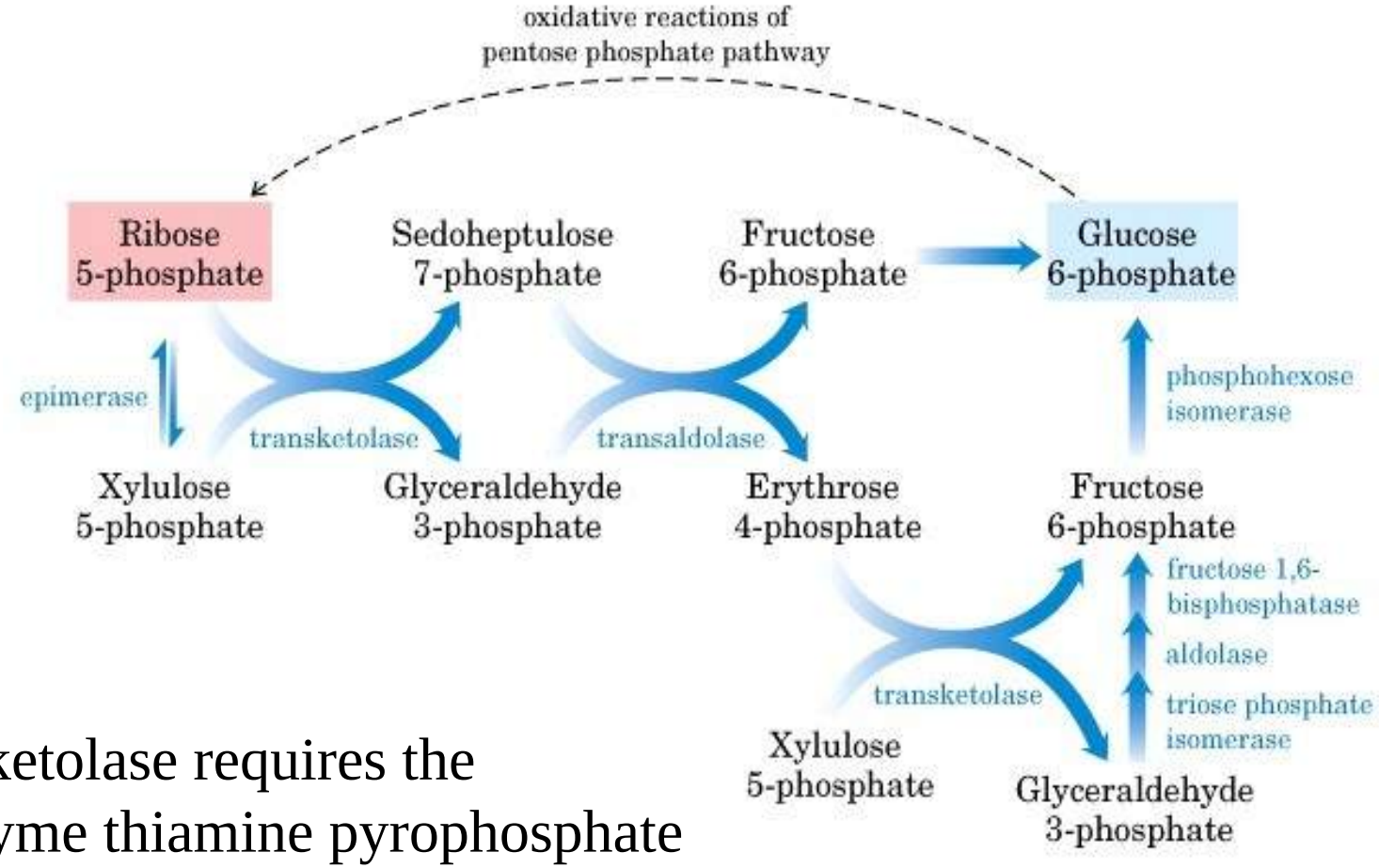
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- For example, many cells that carry out reductive biosynthetic reactions have a greater need for NADPH than for ribose 5-phosphate.

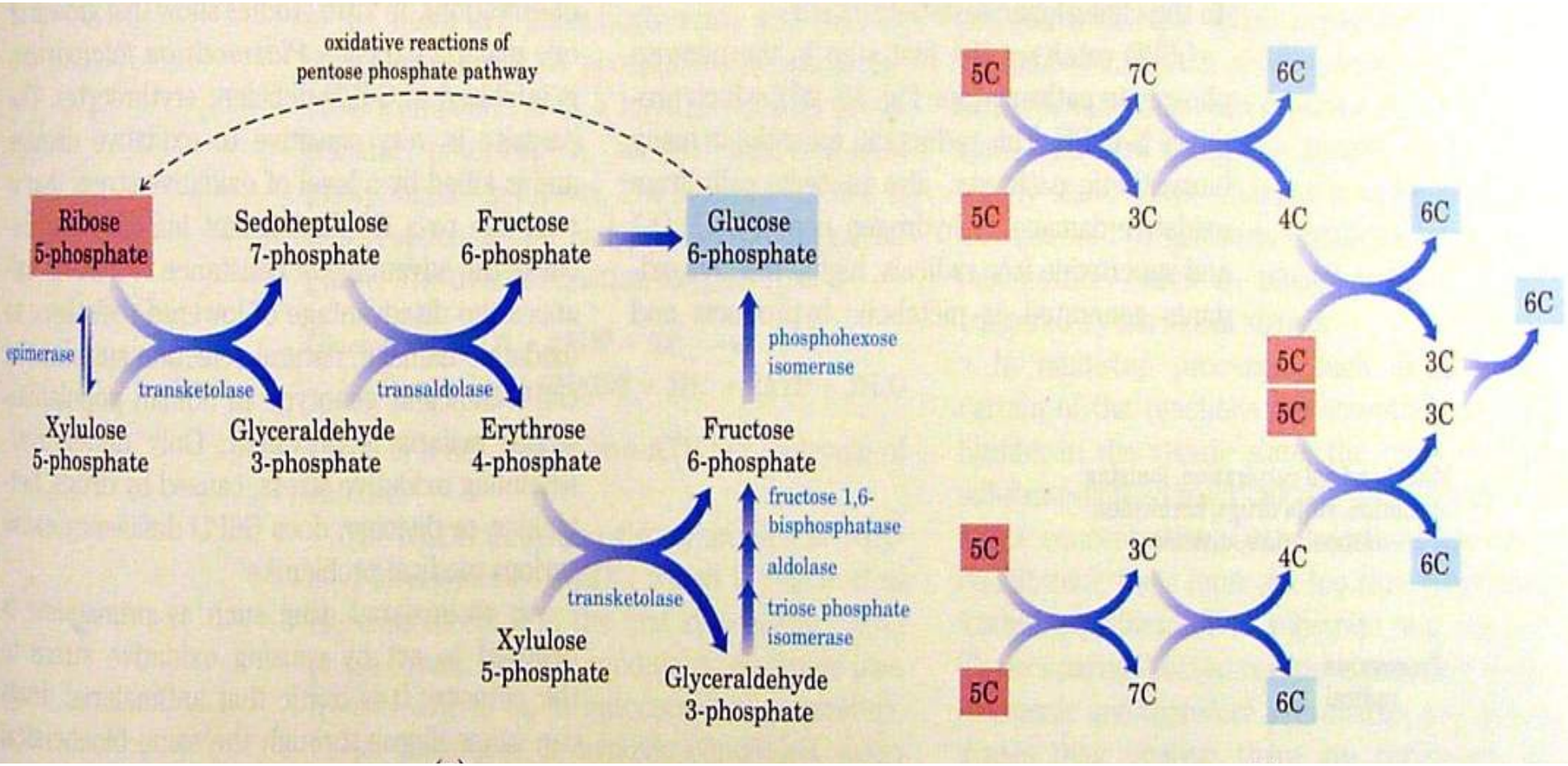
- Transketolase and transaldolase convert the ribulose 5-phosphate produced as an end product of the oxidative reactions to glyceraldehyde 3-phosphate and fructose 6-phosphate, which are glycolytic intermediates.

- In contrast, when the demand for ribose for nucleotides and nucleic acids is greater than the need for NADPH, the nonoxidative reactions can provide the ribose 5-phosphate from glyceraldehyde 3- phosphate and fructose 6-phosphate in the absence of the oxidative steps.

2nd Phase: REVERSIBLE NONOXIDATIVE REACTIONS



Transketolase requires the coenzyme thiamine pyrophosphate (TPP), the transaldolase does not.


oxidative reactions of
pentose phosphate pathway
Ribose
5-phosphate
Sedoheptulose
7-phosphate
Fructose
6-phosphate
Glucose
6-phosphate
epimerase
transketolase
transaldolase
phosphohexose
isomerase
Xylulose
5-phosphate
Glyceraldehyde
3-phosphate
Erythrose
4-phosphate
Fructose
6-phosphate
fructose 1,6-
bisphosphatase
aldolase
transketolase
triose phosphate
isomerase
Xylulose
5-phosphate
Glyceraldehyde
3-phosphate
5C
7C
6C
5C
3C
4C
6C
5C
3C
6C
5C
3C
5C
3C
4C
6C
5C
7C
6C

Regulation of pentose phosphate pathway

- The entry of glucose 6-phosphate into the pentose phosphate pathway is controlled by the cellular concentration of NADPH
- NADPH is a strong inhibitor of glucose 6-phosphate dehydrogenase
- As NADPH is used in various pathways, inhibition is relieved, and the enzyme is accelerated to produce more NADPH
- The synthesis of glucose 6-phosphate dehydrogenase is induced by the increased insulin/glucagon ratio after a high carbohydrate meal.

USES OF NADPH

1. Reductive biosynthesis:

- **NADPH**: a product of the Pentose Phosphate Pathway, functions as a reductant in synthetic pathways, e.g., fatty acid synthesis.

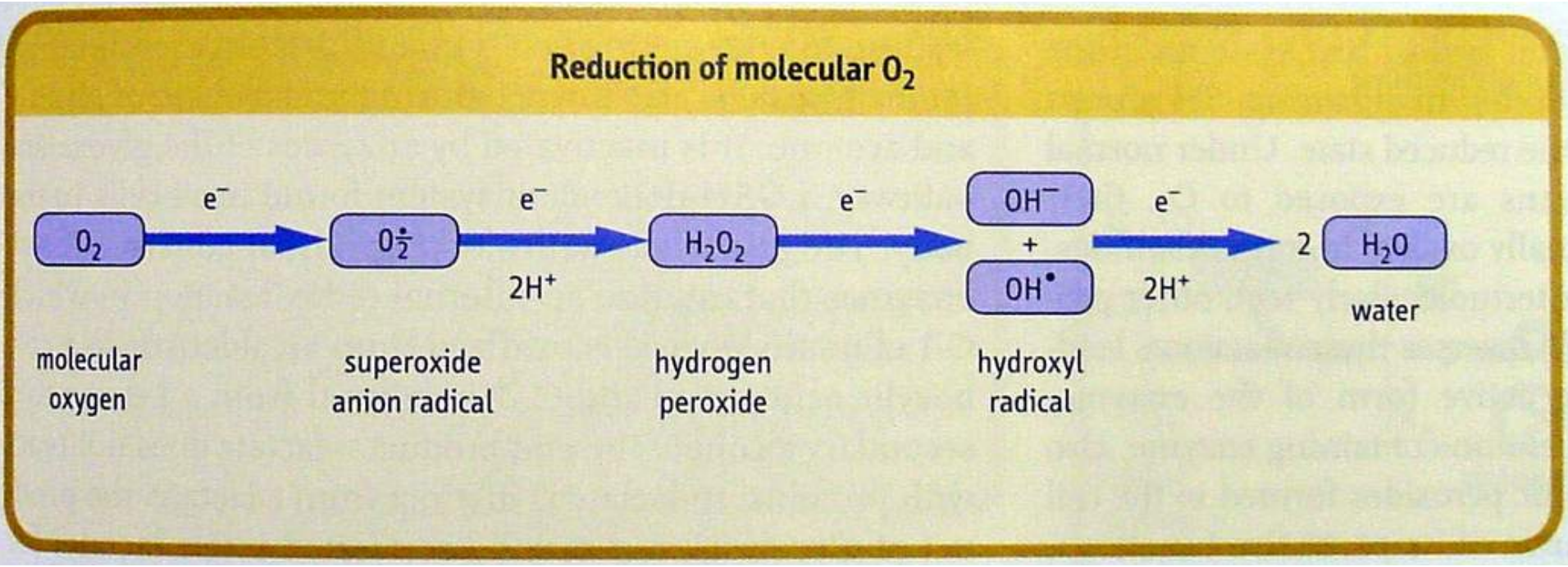
- **NAD^+** serves as electron acceptor

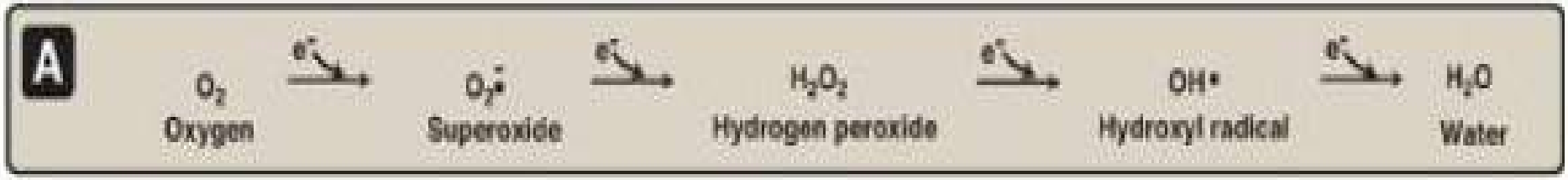
The resultant NADH is reoxidized by the respiratory chain, producing ATP.

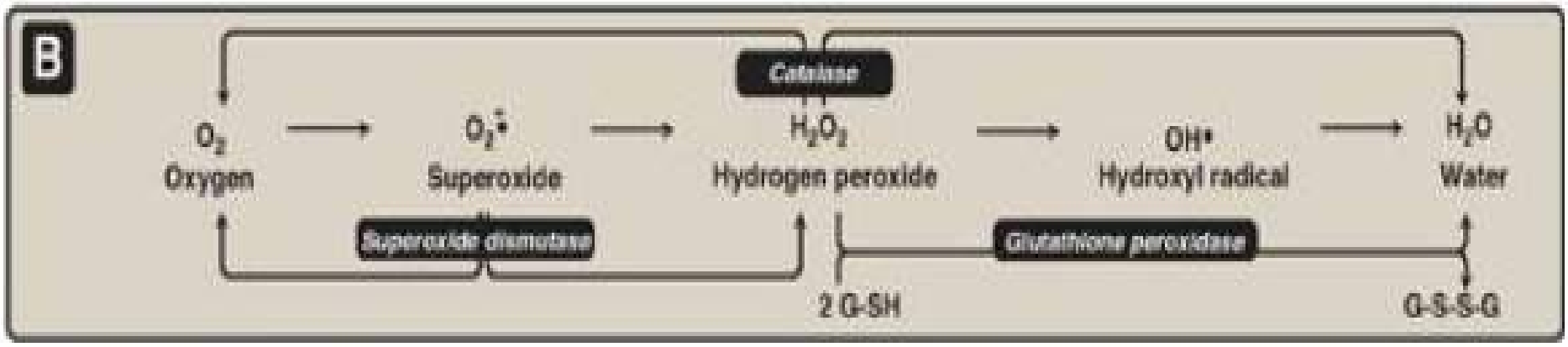
2. Reduction of hydrogen peroxide:

- Pentose phosphate pathway protects cells against reactive oxygen species (ROS).

- Reduction of molecular O_2 in a series of one-electron steps yields superoxide, hydrogen peroxide, hydroxyl radical, and water. The intermediate, activated forms of oxygen are known as reactive oxygen species (ROS).

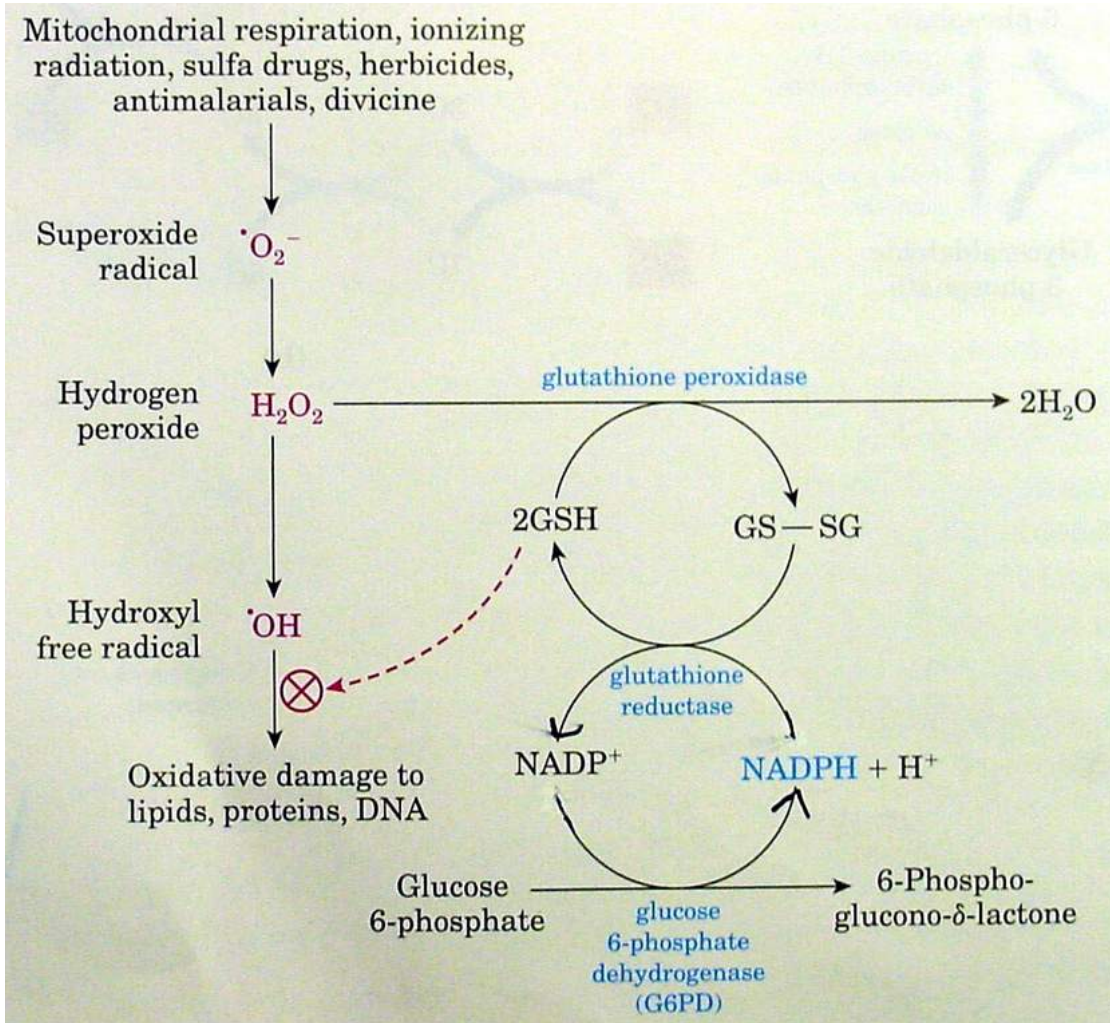

Reduction of molecular O_2
e^-
e^-
e^-
e^-
O_2
$O_2^{\bullet-}$
H_2O_2
OH^-
+
$OH^\bullet$
2
H_2O
$2H^+$
$2H^+$
water
molecular oxygen
superoxide anion radical
hydrogen peroxide
hydroxyl radical


A
O_2
Oxygen
$O_2^{\bullet-}$
Superoxide
H_2O_2
Hydrogen peroxide
$OH^\bullet$
Hydroxyl radical
H_2O
Water


B
Catalase
O_2
Oxygen
$O_2^{\bullet-}$
Superoxide
H_2O_2
Hydrogen peroxide
$OH^\bullet$
Hydroxyl radical
H_2O
Water
Superoxide dismutase
Glutathione peroxidase
2 G-SH
G-S-S-G

Role of NADPH and glutathione in protecting cells against ROS

- **Reduced glutathione (GSH) protects the cell by destroying hydrogen peroxide and hydroxyl free radicals. Regeneration of GSH from it oxidized form (GS-SG) requires the NADPH produced in the glucose 6-phosphate dehydrogenase reaction.**

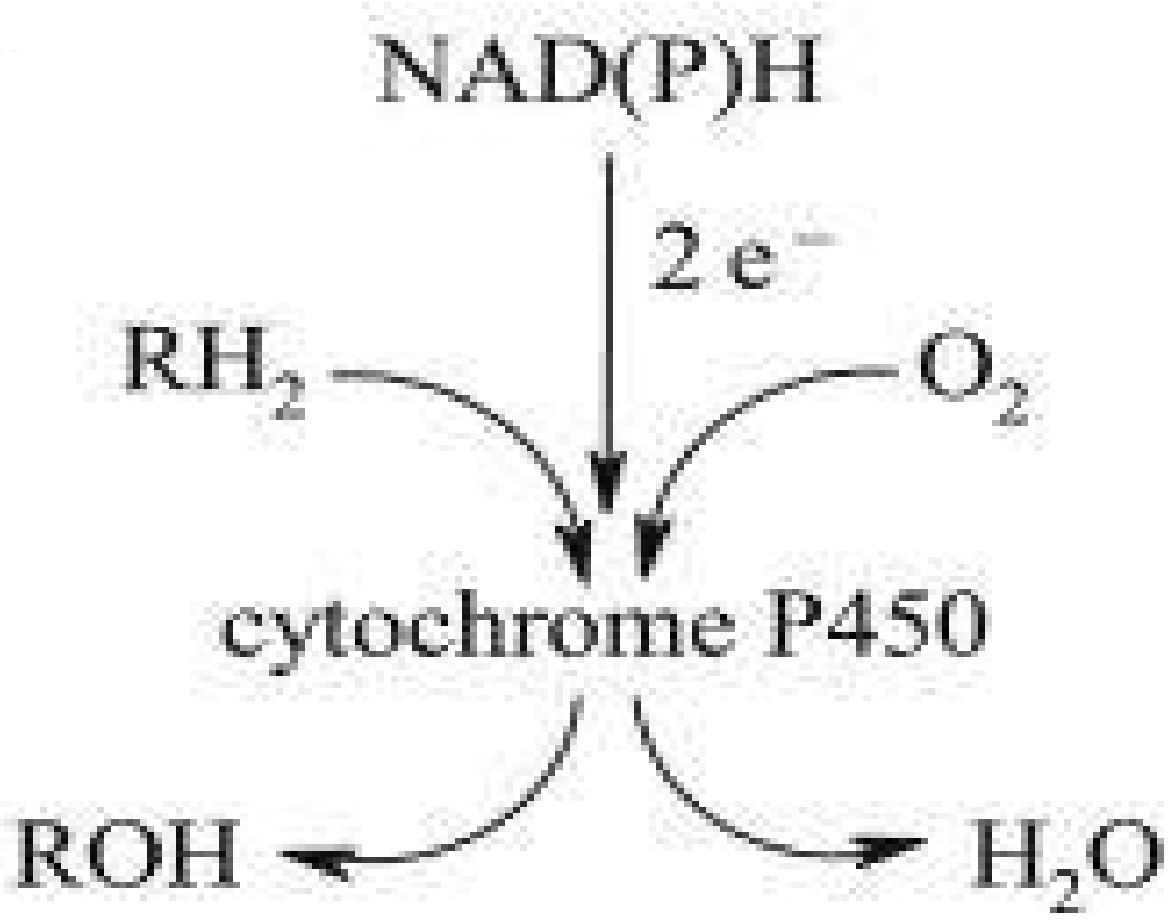


3. Cytochrome P450 monooxygenase system

- Cyto-chrome P450 (CYP) enzymes are heme-containing monooxygenases that participate in a broad variety of reactions.
- In the cytochrome P450 monooxygenase system, NADPH provides the reducing equivalents.

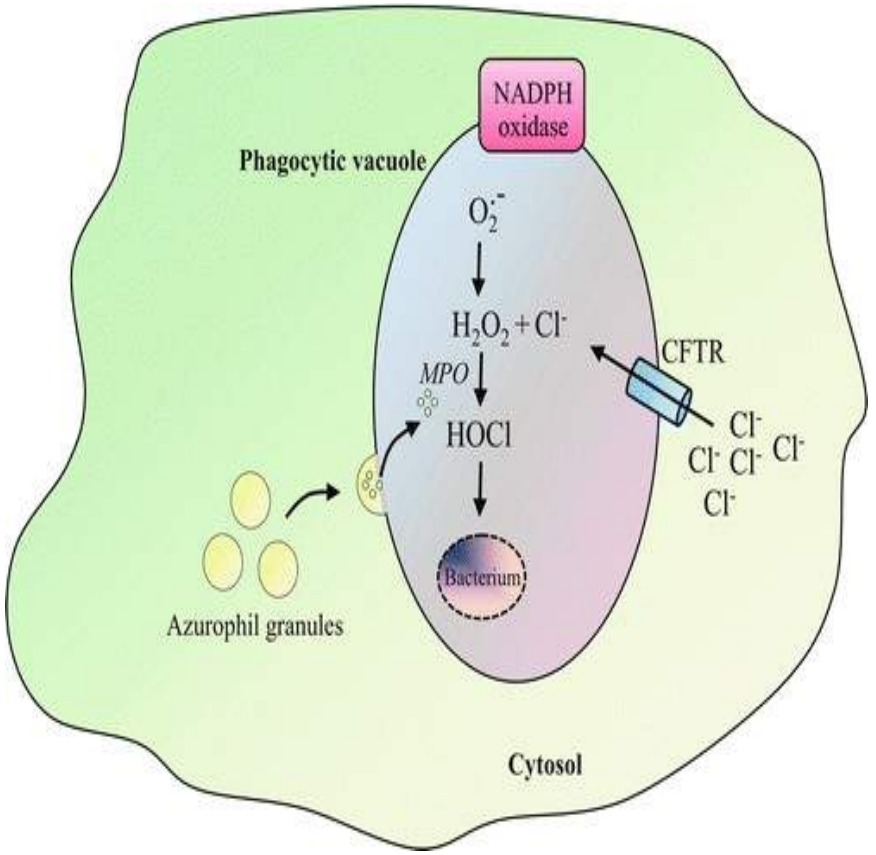
$$R\text{-}H + O_2 + NADPH + H^+ \rightarrow R\text{-}OH + H_2O + NADP^+$$

- This system performs different functions at mitochondrial and microsomal locations in cells.


NAD(P)H
2 e⁻
RH2
O2
cytochrome P450
ROH
H2O

4. Phagocytosis by white blood cells

- Phagocytosis is the ingestion by receptor-mediated endocytosis of microorganisms, foreign particles, and cellular debris by cells such as neutrophils and macrophages (monocytes).
- It is an important defense mechanism, particularly in bacterial infections.



5. Formation of Nitric oxide (NO)

- Nitric oxide synthase is an NADPH - dependant enzyme.
- Nitric oxide is a vasodilator, prevents platelets aggregation.
- Also function as a neurotransmitter and plays an essential role in macrophage bactericidal function.

GLUCOSE 6-PHOSPHATE DEHYDROGENASE DEFICIENCY

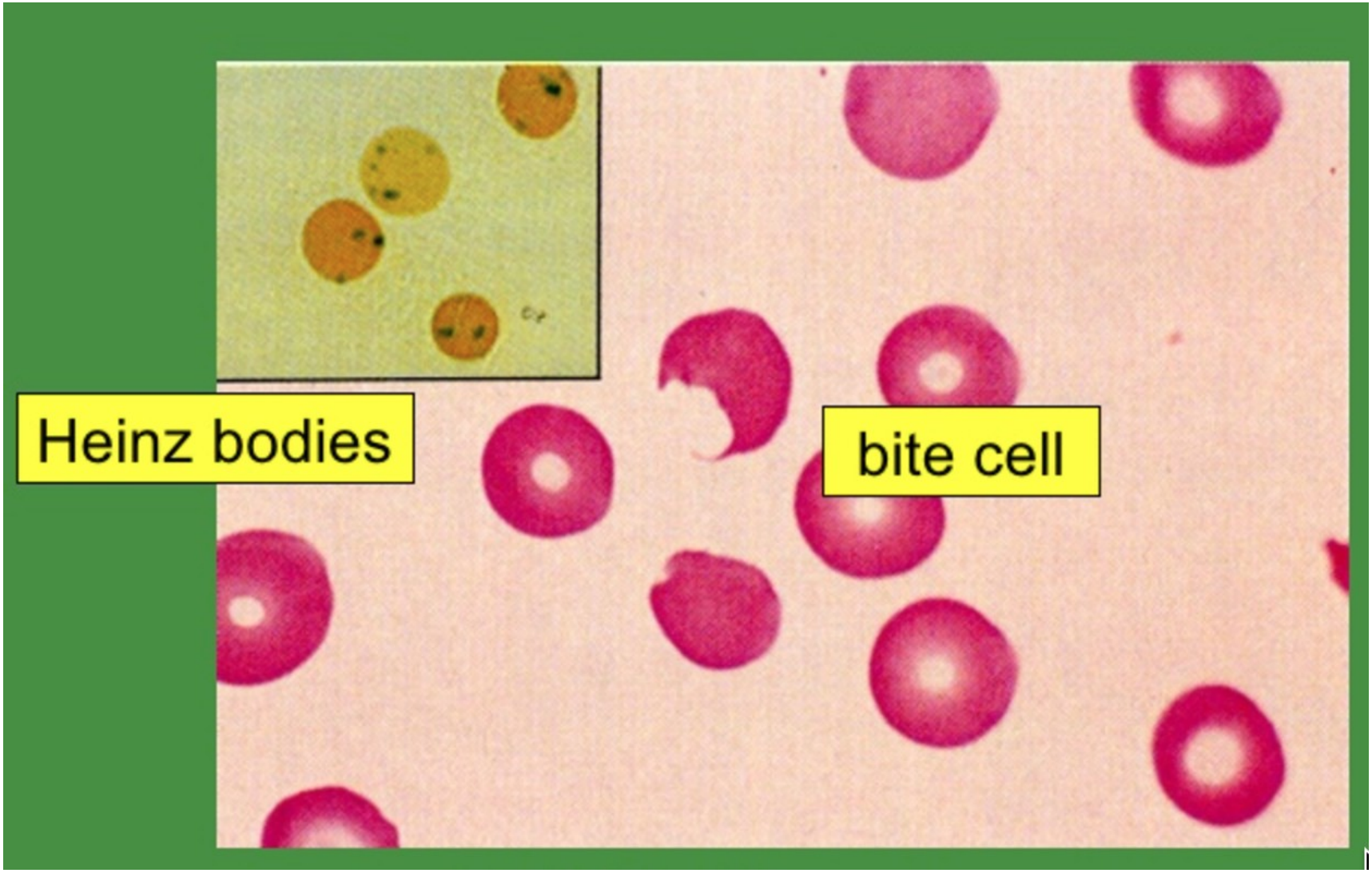
- G6PD deficiency is a X linked hereditary disease characterized by hemolytic anemia caused by the inability to detoxify oxidizing agents.
- G6PD deficiency is the most common disease producing enzyme abnormality in humans.
- It is caused by a number of different mutations in the gene coding for G6PD.
- Only some of the resulting protein variants cause clinical symptoms.

Role of glucose 6-phosphate dehydrogenase in red blood cells

- NADPH is essential for the maintenance of the G-SH pool required for the cellular detoxification of free radicals and peroxides.

- G-SH also helps maintain the reduced states of sulfhydryl groups in proteins, including hemoglobin.

- Oxidation of those sulfhydryl groups leads to the formation of denatured proteins that form insoluble masses (called Heinz bodies) that attach to RBC membranes.


Heinz bodies
bite cell

Precipitating factors in glucose 6-phosphate dehydrogenase deficiency

- Drugs: Antibiotics, antimalarials and antipyretics.

- Foods: Fava beans

- Infection: Infection is the most common precipitating factor of hemolysis in G6PD deficiency. The inflammatory response to infection results in the generation of free radicals in macrophages, which can diffuse into the RBC and cause oxidative damage.

Classification of glucose 6-phosphate dehydrogenase (G6PD) deficiency variants

Class	Clinical symptoms	Residual enzyme activity
I	Very severe (chronic hemolytic anemia)	<10%
II	Severe (episodic hemolytic anemia)	<10%
III	Moderate	10%–60%
IV	None	>60%

THANK YOU